

analysis

Group Members: Ryan and Veronica

Github repository: [View our GitHub repository](#)

```
hallucination_cases <- read.csv("hallucination_cases.csv")
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

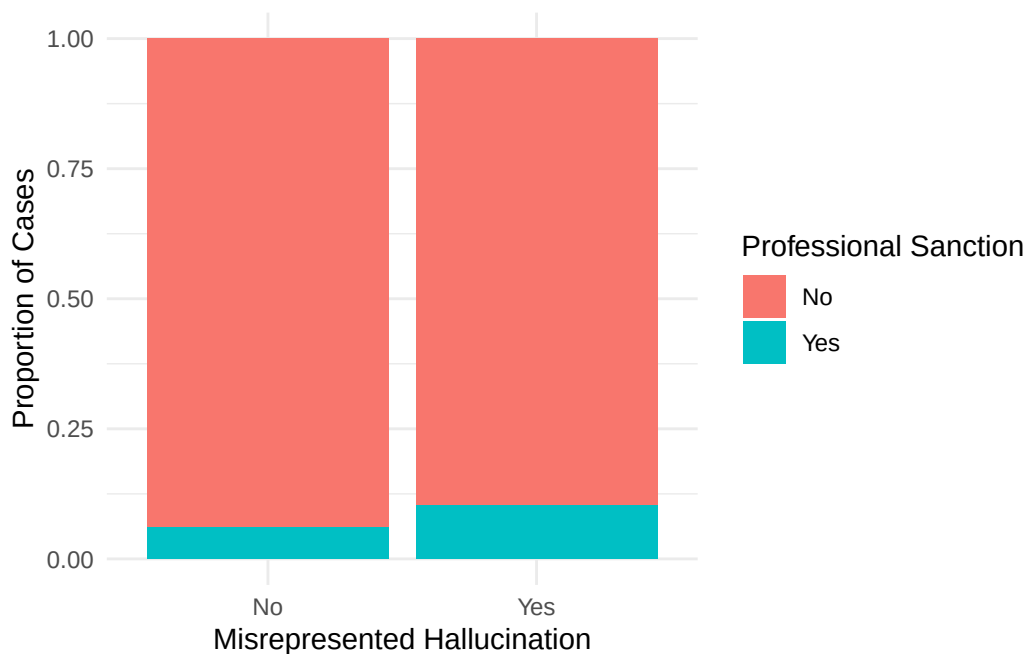
intersect, setdiff, setequal, union

```
library(stringr)
library(tidyr)
library(ggplot2)
```

The **Professional.Sanction** variable describes if a professional sanction, official penalties or disciplinary measures imposed on a licensed worker for an ethical violation, was given. I am curious if having a misrepresented hallucination in the case correlates with an increase in a professional sanction being issued.

```
analysis_data <- hallucination_cases |>
  mutate(
    misrepresented = str_detect(
      `Hallucination.Items`,
      regex("Misrepresented:", ignore_case = TRUE)
    )
  )

ggplot(analysis_data, aes(x = misrepresented, fill = `Professional.Sanction`)) +
  geom_bar(position = "fill") +
  labs(
    x = "Misrepresented Hallucination",
    y = "Proportion of Cases",
    fill = "Professional Sanction"
  ) +
  scale_x_discrete(labels = c("FALSE" = "No", "TRUE" = "Yes")) +
  theme_minimal()
```



This gives the proportion of cases receiving a professional sanction among cases with and without a misrepresented hallucination.

Since Professional Sanction is a binary response variable, we can perform logistic regression.

```
analysis_data <- analysis_data |>
  mutate(
    sanction = if_else(`Professional.Sanction` == "Yes", 1, 0)
  )

model <- glm(
  sanction ~ misrepresented,
  data = analysis_data,
  family = binomial
)

summary(model)
```

Call:

```
glm(formula = sanction ~ misrepresented, family = binomial, data = analysis_data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.7277	0.1200	-22.733	< 2e-16 ***
misrepresentedTRUE	0.5532	0.1640	3.372	0.000746 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1142.7 on 2077 degrees of freedom
 Residual deviance: 1131.2 on 2076 degrees of freedom
 AIC: 1135.2

Number of Fisher Scoring iterations: 5

Because logistic regression coefficients are in log-odds, exponentiating the coefficient gives the odds ratio.

```
exp(0.5532)
```

```
[1] 1.738808
```

The logistic regression estimated that cases with a misrepresented hallucination had 1.74 times the odds of receiving a professional sanction compared with cases without one ($\beta_1 =$

0.5532, $p = 0.000746$). A logistic regression model is appropriate because the response variable, **Professional.Sanction**, is binary, and logistic regression is designed to model the probability of a binary outcome based on the predictor variables. A limitation is that the model only includes whether a case contained a misrepresented hallucination and does not account for other factors such as the severity or number of hallucinations, the type of proceeding, or other potentially significant explanatory variables.